

1. For the following curve, find $\frac{dy}{dx}$ in terms of the parameter:

$$x = (3t^2 - 1)^2$$

$$y = 2t^3$$

2. A curve has the parametric equations $x = t^2$, $y = 1 - \frac{1}{2t}$ for $t > 0$. Find
- The coordinates of the point P where the curve cuts the x -axis;
 - The gradient of the curve at this point and the equation of the tangent to the curve at P;
 - The coordinates of the point where the tangent cuts the y -axis.
3. A curve is given by parametric equations $x = 2t - t^2$, $y = 3t + 4t^2 - 3t^3$.
- Show that it passes through the origin and find all other points where it crosses the x - and y -axes.
 - At points with what coordinates is the curve
 - horizontal;
 - vertical?
4. Find the stationary points on the curve $x = t^2$, $y = t + \frac{1}{t}$.
5. Where does the tangent to the curve $x = \frac{1}{t}$, $y = t^2$ at the point (1,1) meet the curve again?
6. Where does the normal to the curve $x = 6t$, $y = \frac{6}{t}$ at the point where $t = 2$ meet the curve again?
7. A curve has parametric equations $x = t^2$, $y = 2t$.
 R is the point on the curve where $t = r$.
- Show that the normal to the curve at R has gradient $-r$.
 - If S is the point $(s^2, 2s)$, find the gradient of the chord RS in terms of r and s .
 - If the chord RS is normal to the curve at R, show that $r^2 + rs + 2 = 0$.
 - At what point does the normal at the point (9,6) meet the curve again?
8. Sketch the curve given by parametric equations $y = 3\cos t$, $x = 2\sin t$.

The point P has parameter p . Find the gradient of the curve at this point in terms of p .

Find the equation of the tangent to the curve at P and hence find (in terms of p) the coordinates of the points X and Y where this tangent touches the x - and y -axes respectively.

Show the area enclosed by the tangent and the two axes is $6 \operatorname{cosec}(2p)$.

9. Find the equation of the normal to the given curve at the given point and find the coordinates of the point where this normal meets the curve again.
- $x = at$, $y = \frac{c}{t}$ at the point $(ca, \frac{c}{a})$;
 - $x = b(1 - t^2)$, $y = 2b(t - 2)$ at the point $t = b$.

10.

A curve is given parametrically by the equations $x = 4t + 1$, $y = 2t^2$. The point P has the coordinates $(4p + 1, 2p^2)$, with $p \neq 0$.

- (a) Find $\frac{dy}{dx}$ in terms of the parameter t .
- (b) Find the equation of the tangent to the curve at P , and the equation of the normal to the curve at P .
- (c) The point T is the point where the tangent at P cuts the x -axis and the point N is the point where the normal at P cuts the x -axis. Show that the area of triangle TPN is $2p^3(p^2 + 1)$.

11.

A curve is given parametrically by the equations $x = 2a \cos \theta$, $y = 3a \sin \theta$. A point T has coordinates $(2a \cos T, 3a \sin T)$. Show that the area bounded by the x -axis, the y -axis, and the normal to this curve at the point T is $\frac{25}{24}a^2 \sin 2T$.

12. If a parabola is defined parametrically by the equations $x = at^2$ and $y = 2at$ show for any a that the distance from the parabola to the line $x - y + 4a = 0$ is at its least when $t = 1$, and show that this shortest distance is $\frac{3}{2}\sqrt{2a}$.